

AMENI Khabthani

AI Engineer | Voice AI Specialist

📍 Mahdia, Tunisia 📞 +216 23 513 019 ✉️ ameni.khabthani2@gmail.com
🌐 [linkedin.com/in/amenikhabthani](https://www.linkedin.com/in/amenikhabthani) 🐙 github.com/Amenikhabthani

SUMMARY

AI Engineer with hands-on experience building and deploying production-grade AI solutions for international clients, including NLP, LLM-based, and speech systems and conversational AI assistants. Experienced across the full lifecycle, from data preparation and model fine-tuning to API development and cloud deployment.

SKILLS

AI & LLMs: Machine Learning, Deep Learning, LLMs, Generative AI, AI Agents, Prompt Engineering, Fine-Tuning (PEFT, QLoRA)
Speech & Voice AI: ASR, TTS, VAD, Speaker Diarization, OpenAI Whisper, Coqui XTTS v2, Orpheus TTS
Tools & Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, LangChain, FastAPI, Flask, Supabase, Docker, Cloud Services, Git, CI/CD pipelines

EXPERIENCE

Freelance AI Engineer

Feb 2025 – Present

Freelance | Remote

- Custom Text-to-Speech Systems:** Fine-tuned Coqui XTTS v2 (multilingual text-to-speech) models for multiple clients across languages, accents, and voice types, including Hindi, Hinglish, Moroccan Arabic, a female Saudi voice, and children's voices; supported each project from data preparation to training and evaluation.
- Multilingual Dubbing Automation:** Transformed a third-party dubbing workflow into a custom multilingual pipeline using Whisper Large v3 ASR, GPT-4 translation, speaker diarization, and hybrid XTTS/Orpheus TTS, deployed via FastAPI on Azure App Service.
- AI Travel Agent:** Engineered a FastAPI voice assistant for real-time hotel search and booking, integrating Whisper ASR, Groq LLMs, and TTS with intent detection, entity extraction, and conversational memory.
- AI-Based Astrology App:** Designed and deployed an end-to-end API using LLM prompt engineering and intent classification for real-time personalized responses, with Supabase for user data.
- Medical Translation System:** Built an English–Arabic text-to-text translation system that preserves domain-specific medical terminology with high clinical and semantic accuracy, for healthcare transcription tools.

Data Science Intern

Mar 2024 – Aug 2024

Satoripop | Sousse, Tunisia

- Fine-tuned Whisper for French, English, and Tunisian dialect ASR, reducing word error rate (WER) by 45% on the target datasets.
- Fine-tuned XTTS v2 for speech synthesis, improving naturalness and intelligibility, validated through user feedback.
- Built FastAPI APIs that decreased response times by 30% and integrated with external applications.
- Developed Gradio demo interfaces to improve accessibility and simplify manual testing.

Data Science Intern

Jun 2023 – Aug 2023

Stunas | Sousse, Tunisia

- Developed an OCR invoice label recognition model using DocTR for text detection and Named Entity Recognition (NER) for label extraction, reducing manual input time by 50%.

EDUCATION

Master's Degree in Data Science, AI, and HealthTech

Oct 2022 – Oct 2024

Pristini School of AI

Bachelor's Degree in Electrical Engineering (Automation and Industrial Computing)

Sep 2019 – Jun 2022

ISSET Mahdia

CERTIFICATIONS

Coursera: [Natural Language Processing Specialization](#); [Generative AI with Large Language Models](#); [Generative AI Engineering with LLMs Specialization](#); [DeepLearning.AI Specialization](#); [Python Data Structures](#); [Data Analysis with Python](#)

AWS Academy: [Cloud Foundations](#)

LANGUAGES

Arabic (Native) | English (Professional) | French (Intermediate)